

Karthik Reddy Yalala

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EDUCATION

Arizona State University

Tempe, AZ

Bachelor's in Computer Science (GPA: 4.0)

May 2027

- Relevant Coursework: Algorithms, Data Structures, OOP, Operating Systems, Software Engineering, Cybersecurity, Artificial Intelligence, Machine Learning, Web Development, Discrete Mathematics
- Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, Java, HTML/CSS

Software Engineering: System Architecture, REST APIs, Unit Testing, Feature Testing, Code Review, Git, GitHub, Agile, CI/CD

AI/ML & GenAI: Prompt Engineering, RAG, Claude, OpenAI APIs, Amazon Bedrock, Bedrock Guardrails, AgentCore, LangChain, LangGraph, CrewAI, PyTorch, TensorFlow, Hugging Face

Web & APIs: React.js, Node.js, FastAPI, Express.js, REST APIs, WebSocket, SSE, Tailwind CSS

Cloud Infrastructure (AWS): Lambda, API Gateway, CloudFront, S3, DynamoDB, WAF, Cognito, Azure AD, Docker

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Firebase, SQL

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02)

PROFESSIONAL EXPERIENCE

AI Enablement Intern

May 2026 – Aug 2026

Avis Budget Group (Fortune 500)

Parsippany, NJ

- Spearheaded **4 end-to-end software initiatives** from system architecture and API design to front-end and back-end implementation, presenting designs to the Director and VP of Customer Experience and demoing the platform directly to the **CEO**; saved **170+ engineering and product hours** across projects
- Engineered a multilingual RAG knowledge search engine on AWS (Lambda, API Gateway, Bedrock, DynamoDB, CloudFront) with brand-aware EN/FR/ES retrieval, validated end-to-end with unit and adversarial test suites; positioned to serve **millions of customers globally** and drive **\$1M+ annual cost savings**
- Designed and deployed an internal AI assistant on Amazon Bedrock AgentCore with multi-turn session memory, query rewriting, real-time SSE streaming, Azure AD SSO/MFA, and a zero-downtime draft-test-approve-publish config pipeline with two-layer prompt safety scanning
- Built a natural-language-to-SQL analytics engine with well-commented, production-ready code, reducing data turnaround **from days to seconds** and saving **50 engineering and 20 product hours**; enforced schema-level PII guardrails and validated composite joins on production data
- Launched an AI-powered Jira delivery assistant using Rovo AI workflows deployed to production for organization-wide adoption; delivered AI visibility research to the VP of Customer Experience

Data Analyst Intern

May 2025 – July 2025

Food Forest AI

Philadelphia, PA

- Engineered GPT-powered data extraction pipelines to scrape, parse, and validate **200+ company records** against agreed schemas, improving dataset quality by **40%** for an AI-driven B2B search platform
- Automated deduplication and anomaly flagging via Google Sheets, eliminating manual data cleaning and ensuring clean structured output for downstream AI ingestion

Software Engineer Intern

May 2024 – July 2024

YJR Realtors

Hyderabad, India

- Designed and coded a responsive customer-facing interface with AI-powered chat agents to automate end-to-end property inquiry resolution; developed a Python SendGrid automation script at scale, eliminating hours of manual outreach weekly
- Built an internal lead management dashboard with automated follow-up logic, real-time status tracking, and bug-free delivery through iterative code review, improving team response time by **40%**

TECHNICAL PROJECTS & ACHIEVEMENTS

Interviewer.ai | LangGraph, Amazon Bedrock, FastAPI, React, DynamoDB, AWS Lambda

2026

- Architected a **five-agent LangGraph pipeline** on AWS with schema-validated inter-agent contracts, system architecture documentation, front-end and back-end implementation, and per-agent evaluation harnesses for SDE and AI Engineer roles
- Engineered cross-session adaptive memory with DynamoDB; deployed full-stack serverless system (React on CloudFront/S3, FastAPI on Lambda behind API Gateway) with Cognito auth and complete unit testing coverage

Deep Research Agent Team | CrewAI, LangGraph, OpenAI APIs, Pinecone

June 2025

- Built a modular multi-agent research system with planner-executor-validator architecture, retry logic, exponential backoff, and output caching, reducing manual research effort by **70%**